(In)Efficient Markets

Jinfan Zhang

- Reference: Chapter 11, BKM

Overview

- Define what we mean by *market efficiency*
- Review the efficient markets debate
- Look at evidence for and against market efficiency
- Discuss rational and behavioral perspectives on market anomalies
- Look at limits to arbitrage

Winners of 2013 Nobel Prize for Economics

Press Release

14 October 2013

The Royal Swedish Academy of Sciences has decided to award The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel for 2013 to

Eugene F. Fama
University of Chicago, IL, USA

Lars Peter Hansen
University of Chicago, IL, USA

and

Robert J. Shiller
Yale University, New Haven, CT, USA

"for their empirical analysis of asset prices".

Bold claims in Media

The New Yorker:

INEFFICIENT MARKETS: A NOBEL FOR SHILLER (AND FAMA)

Business Insider:

Robert Shiller Debunks Eugene Fama's Ideas To His Face During This Nobel Prize Presentation

Forbes

What Eugene Fama Could Learn From His Fellow Nobel Winner Robert Shiller

- Review of the Market Efficiency Debate

Efficient Market Hypothesis (EMH)

- EMH says stock prices already reflect all available information
- A forecast about favorable *future* performance leads to favorable *current* performance, as market participants rush to trade on new information.
- Result: Prices change until expected returns are exactly commensurate with risk.

Definition of market efficiency

- Market efficiency means that security prices are right
  - i.e. they fully reflect all available information

- Equivalently, all prices equal their fundamental value

$$FV_0 = \frac{E(CF_1)}{1+r} + \frac{E(CF_2)}{(1+r)^2} + \ldots$$

  - use all available information to make the best possible cash-flow forecast
  - use a discount rate *r* that is appropriate for the risk

Versions of the EMH

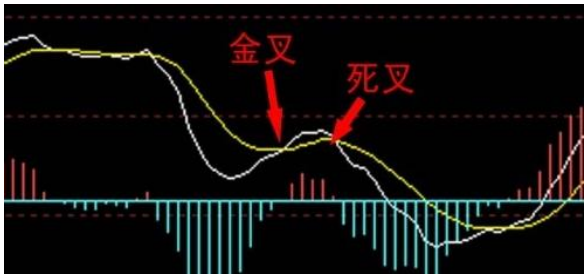
- Weak
  - Market trading data
- Semi-strong
  - All public information
- Strong
  - All information, including insiders' private information
- All versions assert that prices should reflect *available* information

Implications of the EMH(weak form)

- Using prices and volume information to predict future prices (Technical Analysis)
- Success depends on a sluggish response of stock prices to fundamental supply-and-demand factors.
- Test of the weak form efficiency

Technical Analysis

- Candle Plot
  - Gold Crossing (金叉)
    - Short term moving average line surpass the long term moving average line upward.
  - Dead Crossing (死叉)
    - Short term moving average line surpass the long term moving average line downward.

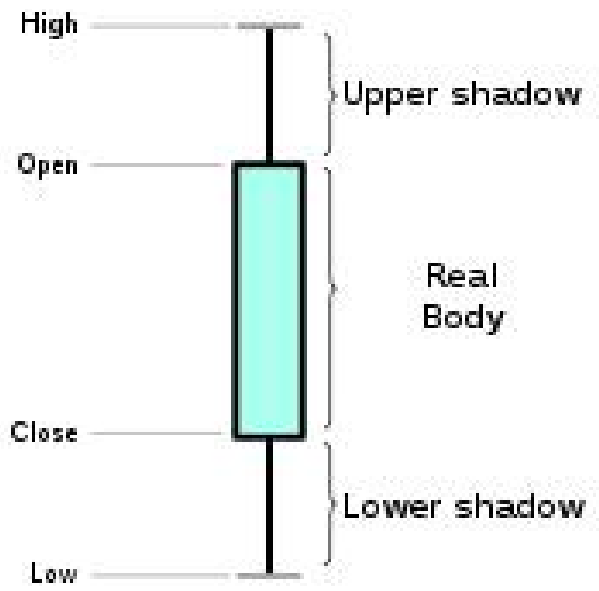


白线：short window [more volatile]

黄线：long window (last 30-days)



Candle plot





- **白三兵**
  - 白三兵是牛市的逆转模式，形成三个连贯的长白蜡烛。在一段走低后，白三兵模式指示市场心态的改变和从熊市到牛市的逆转趋势。牛市的确认勿庸置疑，有时逆转会形成一个价格支撑位。
- **三个黑乌鸦**
  - 熊市逆转模式，由三个连续的黑烛身组成。每天开盘时高于昨天的最低价，但收盘时低于昨天的最低价。

Implications of the EMH (semi-strong form)

- Fundamental Analysis - using economic and accounting information to predict stock prices
  - Try to find firms that are better than everyone else's estimate.
  - Try to find poorly run firms that are not as bad as the market thinks.
  - Test of the semi strong form efficiency

Never generate α. (mis-pricing)

Only way to generate money is exposure to risk

Implications on Asset Management

- Active Management
  - An expensive strategy
- Passive Management:
  - Accept EMH, don't attempt to outsmart the market
  - Index Funds and ETFs with very low costs
- The role of portfolio management in an efficient market
  - Diversification
  - Appropriate risk level

Implications of market efficiency

- Prices react to *new* information quickly and to the right extent
- There is no free lunch:
  - the only way to get higher returns is by taking on more risk
  - there is no information out there that can be used to construct strategies that earn returns higher than required for their risk
    - A logic problem

Theoretical argument for market efficiency

- Milton Friedman (1953)
  - If a security becomes mispriced, smart investors should quickly take advantage of it, and the mispricing should disappear quickly
    - this process is often called “arbitrage”
  - Anyone who claims markets are inefficient, must explain why the arbitrage process breaks down
    - major challenge for behavioral finance

Why do we care?

- In an efficient market, resources have been allocated efficiently by markets.
- There is no role for intervention
- In an inefficient market, capital is not properly allocated across firms.
- Potentially, there is a role for regulation

What does the evidence suggest?

How can we tell if markets are inefficient?

- Can look for stocks whose prices seem wrong
  - but hard to be sure the price is wrong
  - when we say 'prices are wrong', we are **implicitly stating what 'correct' is**
- Can look to see if new information is quickly and fully incorporated into prices
- Can look for investment strategies that seem to earn higher average returns than they deserve for their risk
  - but hard to judge their risk, so it is hard to be sure they really are earning more than they deserve

Evidence for market efficiency

- Professional money managers do not beat the market on average
- What does this mean?
  - There are no market inefficiencies to exploit?

Evidence for market efficiency

- New information appears to be quickly incorporated into prices
  - e.g. suppose “news” is the announcement of a takeover
  - look at the target firm’s stock price reaction to the news
  - average over many companies to see the typical pattern
  - This is called an “event study”

Event Studies

- Empirical financial research enables us to assess the impact of a particular event on a firm's stock price.

- The **abnormal** return due to the event is the difference between the stock's actual return and an estimate of the stock's return **in the absence of the event**.

Event Studies

- Expected Return

  $r_t = a + br_{Mt} + e_t$

- Abnormal Return = (Actual - Expected)

  $e_t = r_t - (a + b*r_{Mt})$

Example: Cumulative Abnormal Returns Before Takeover Attempts: Target Companies

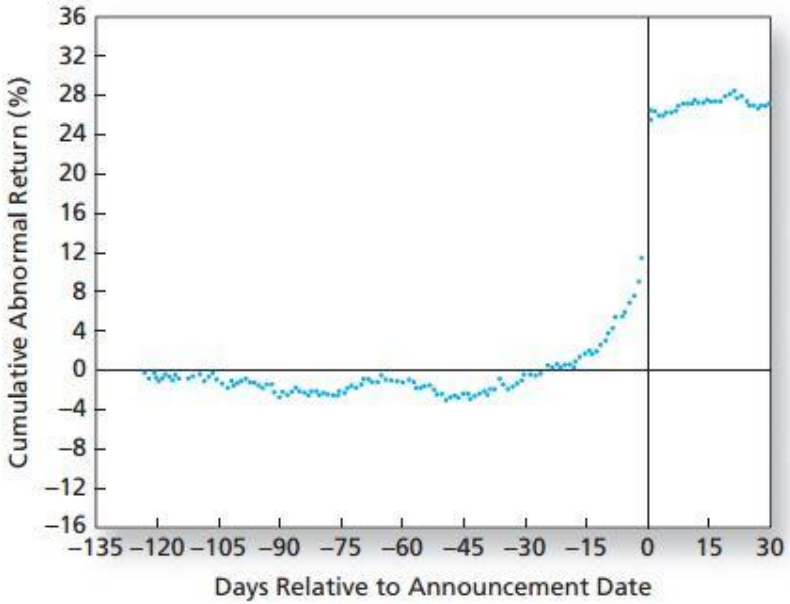
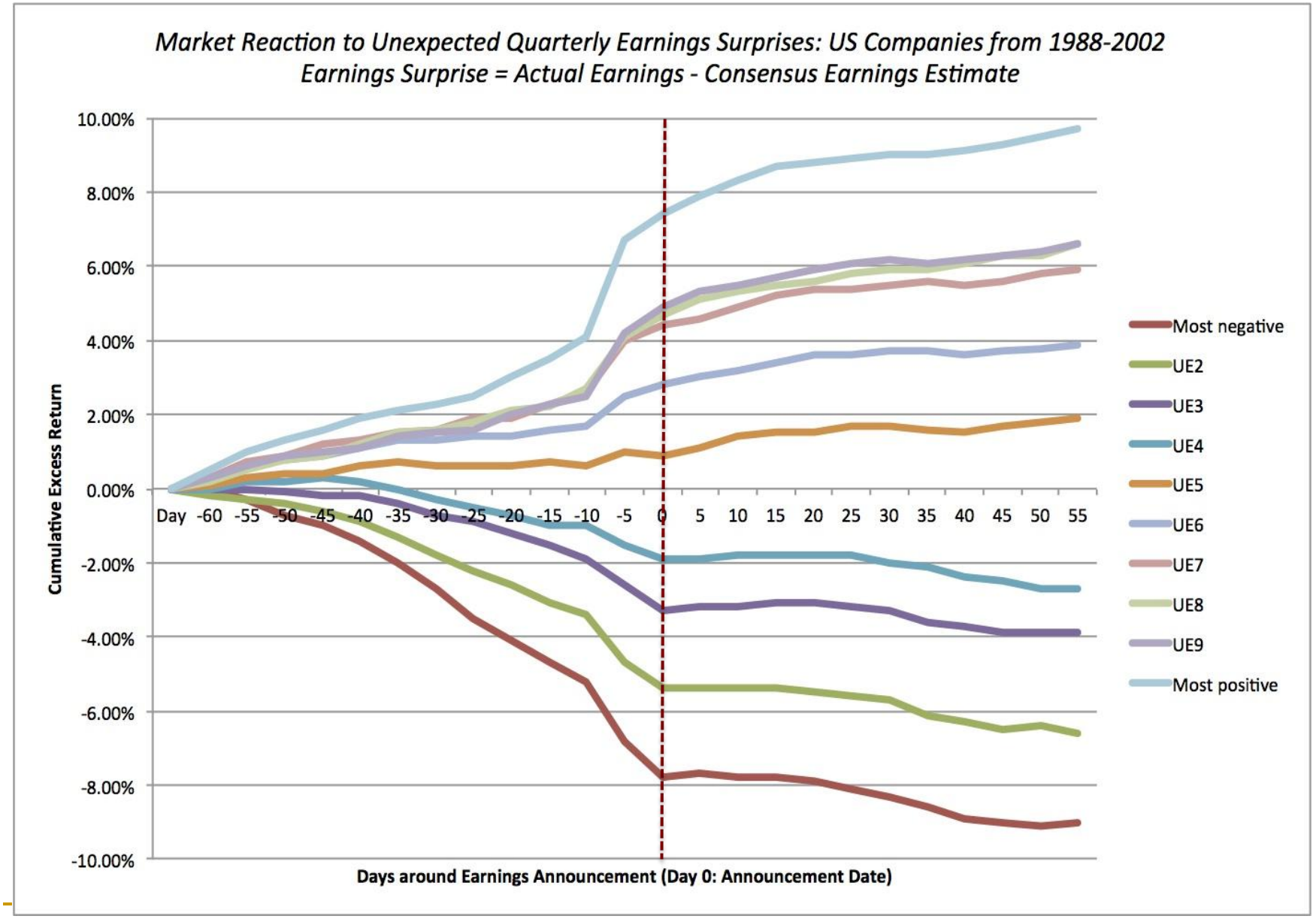


Figure 11.1 Cumulative abnormal returns before takeover attempts: target companies

Source: Arthur Keown and John Pinkerton, "Merger Announcements and Insider Trading Activity," *Journal of Finance* 36 (September 1981). Used with permission of John Wiley and Sons, via Copyright Clearance Center. Updates courtesy of Jinghua Yan.

Slow incorporation of news into stock prices

- Earnings news does not appear to be quickly and fully incorporated into prices
- On unexpectedly good news, a firm's stock price jumps up, but continues to drift up in the weeks thereafter
- On unexpectedly bad news, a firm's stock drops in value, but continues to drift down in the weeks thereafter


Market Reaction to Unexpected Quarterly Earnings Surprises: US Companies from 1988-2002
Earnings Surprise = Actual Earnings - Consensus Earnings Estimate
10.00%
8.00%
6.00%
4.00%
2.00%
0.00%
-2.00%
-4.00%
-6.00%
-8.00%
-10.00%
Cumulative Excess Return
Day -60 -55 -50 -45 -40 -35 -30 -25 -20 -15 -10 -5 0 5 10 15 20 25 30 35 40 45 50 55
Days around Earnings Announcement (Day 0: Announcement Date)
Most negative
UE2
UE3
UE4
UE5
UE6
UE7
UE8
UE9
Most positive

Can the market calculate addition and subtraction?

- Huancheng Du, Xiaoran Ni, Jinfan Zhang, The “Ex-dividend Day” Anomaly under a Behavioral Dividend Clientele View: Evidence from China

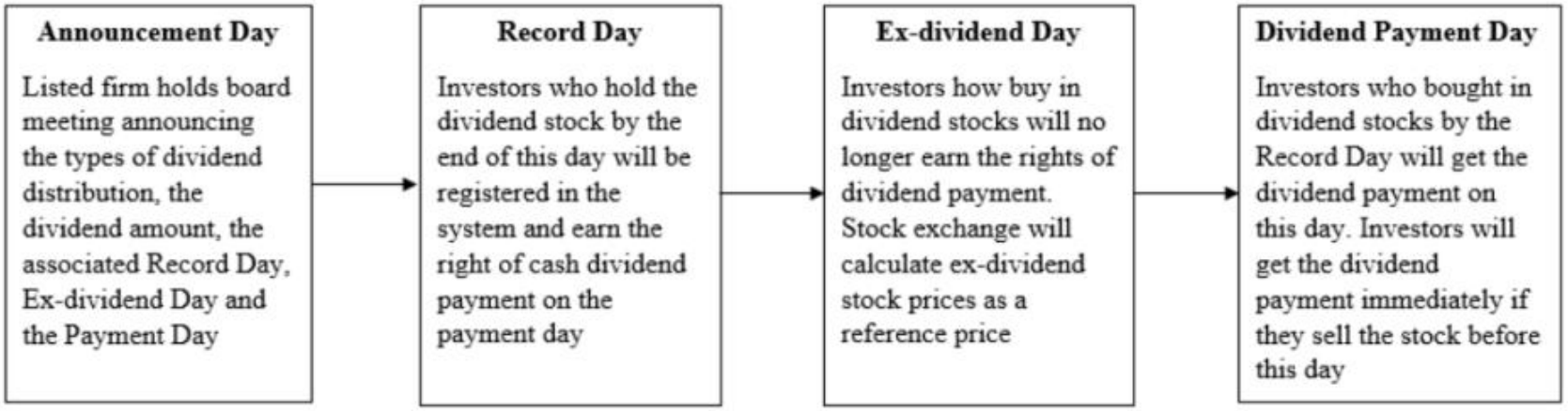


Figure 1. **Timeline for Dividend Distribution**

This figure presents the four important dates of cash dividend distribution process. We provide brief descriptions for each of these four dates.

Can the market calculate addition and subtraction?

- No

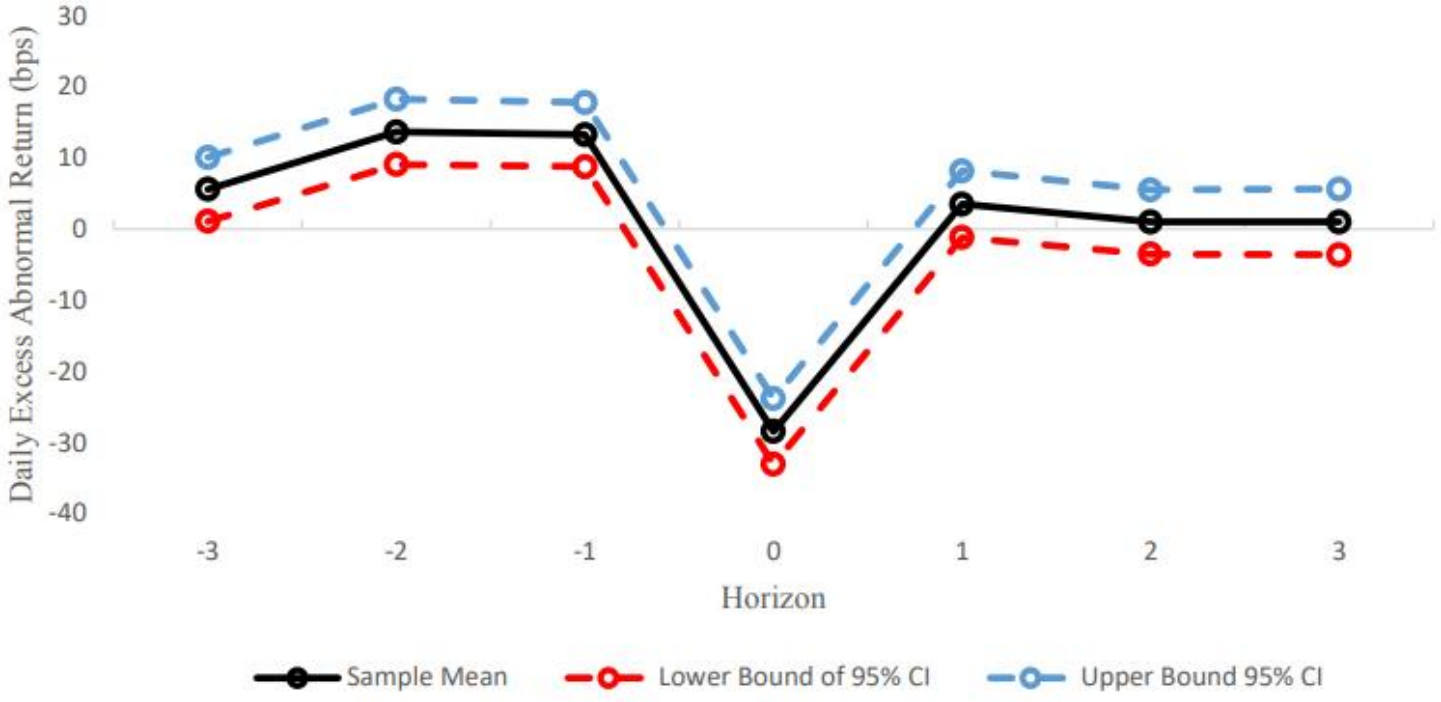


Figure 3 (a). **Daily Excess CAPM Adjusted Return Around Ex-dividend Day**

Can the market calculate addition and subtraction?

- No.
- Why? Naïve investors think divided as free lunch

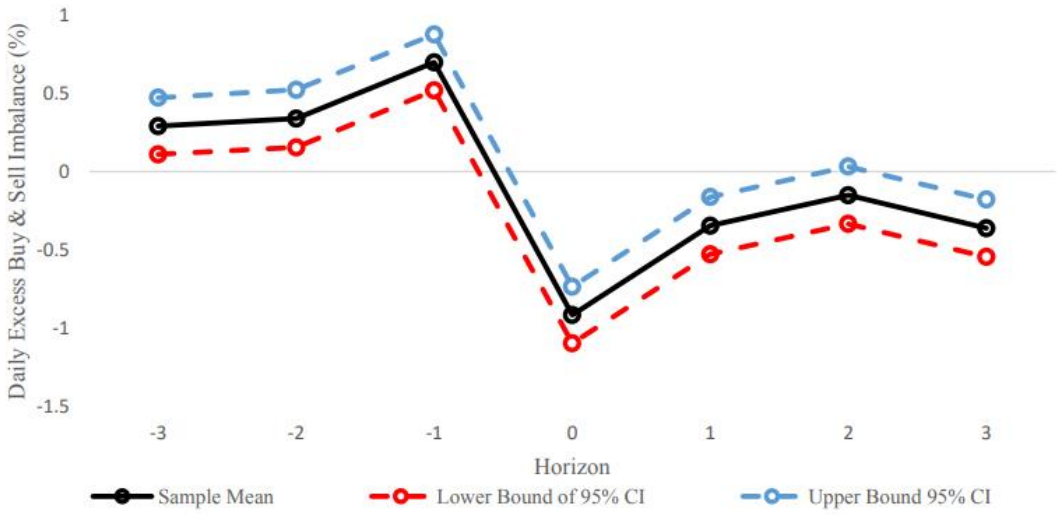


Figure 4 (c). **Daily Excess Buy-Sell Imbalance for Small Investor**

This figure plots the sample mean of the daily excess buy-sell imbalance for small retail investors. The lower bound and upper bound of 95% confidence interval are shown as the red and blue dashed lines.

Can we come up with a strategy that makes money?

Investment strategies that seem to beat the market

- These are strategies that seem to earn higher average returns than they should, for their risk

- e.g. if you regress excess strategy returns on excess market returns, you get a positive alpha intercept:

$$r_{P,t} - r_{f,t} = \alpha_P + \beta_P(r_{M,t} - r_{f,t}) + \epsilon_{P,t}$$

Average Annual Return for 10 Size-Based Portfolios, 1926 – 2011

(mid-cap strategy:
balance risk for bankrupcy & low return)

沪深300 大

中证500 中 → higher return.



Figure 11.3 Average annual return for 10 size-based portfolios, 1926–2011

Source: Authors' calculations, using data obtained from Professor Ken French's data library at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

the difference can NOT be fully explained by beta

指数增强 ETF

Example: Value vs. Growth

Value premium strategy.

MKT. SMB. HML

- Form portfolios based on prices to fundamentals
  - use price to earnings (P/E) ratio
  - could also look at price to book (P/B) and price to cash-flow (P/CF) ratios

"cheap stock" low PB or PE

- Over many decades, average return of value stocks is much higher than that of growth stocks

"expensive stock".

high PB or PE → high growth rate.

  - but doesn't appear to be riskier
- Value stocks (with low ratios) appear to earn higher returns than they "deserve," for their risk
- Growth stocks (with high ratios) appear to earn lower returns than they "deserve," for their risk

Average Return as a Function of Book-To-Market Ratio, 1926–2011



what risk is these "cheap" stocks bearing? Argue.

Figure 11.4 Average return as a function of book-to-market ratio, 1926–2011

Source: Authors' calculations, using data obtained from Professor Ken French's data library at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Reactions to the evidence

- Behavioral finance people say:
  - these strategies earn more average return than they should, for their risk
  - they must therefore be exploiting mispricing

- Rational finance people say:
  - the strategies are risky, but the standard measures of risk fail to capture this
  - look for new ways to measure risk under which the strategies do look risky

- We can directly check whether stock price is wrong
  - Of course, the challenge is we are not sure what is the correct price

Evidence against market efficiency

- Cases where prices seem to move even though there is no new information
  - e.g. the 20% drop in the stock market in one day in October 1987
  - The Treasury "flash" rally, occurred on Oct. 15, 2014 and only lasted for about an hour. But the ferocity of the ascent — and the simultaneous eye-popping plunge in the yield — prompted widespread concern about the functioning of the market for Treasury securities.
- There are investment strategies that seem to have earned higher average returns than is consistent with their risk ("beat the market")
  - these are so called "anomalies"
- Shiller (1981) provided evidence that the stock market prices are too volatile to be explained by changes in fundamentals.

$$P = \frac{D_{t+1}}{(r_f + r_{premium}) - g}$$

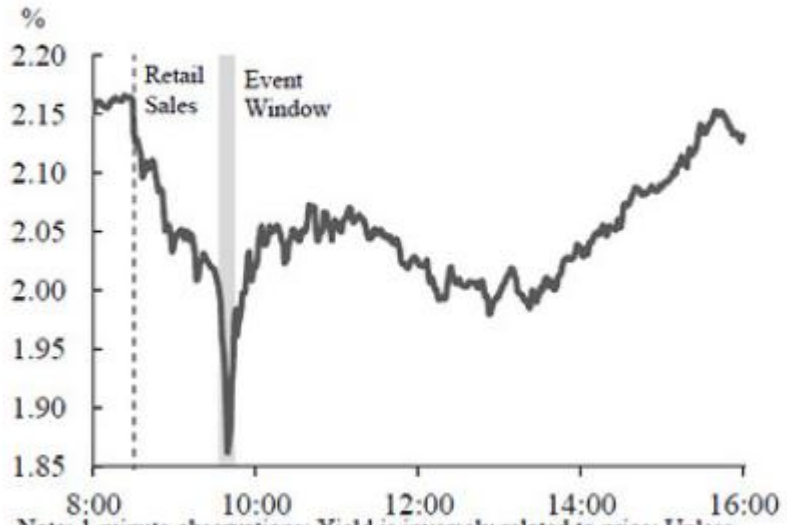
cause volatile?

- stable
- not cause that much volatile in P

Shiller argues that. (people sometimes ask large premium and sometimes small).

Specific Examples--The Oct. 15, 2014 "flash rally" in US Treasury bond market

Figure 2.1: 10-Year Treasury Yield on October 15 (Cash)



Note: 1-minute observations; Yield is inversely related to price; Unless otherwise noted, intraday figures show October 15 US trading hours
Source: Staff calculations, based on data from Bloomberg.

price ↑ yield ↓

Typically, e.g. 911, disaster make people panic.
sell stocks to buy treasury bond.



Joint Staff Report:

THE U.S. TREASURY MARKET ON OCTOBER 15, 2014

The report asserts that the banks and principal trading firms took steps to protect themselves from a rise in Treasury prices — and those steps seemed to drive the prices even higher. The banks widened the gap at which they would buy and sell Treasuries, a standard practice in volatile moments in the markets. At one point, they stopped offering to sell Treasuries. The principal trading firms, for their part, reduced the average size of their trades. "Both actions served as risk management strategies by reducing the number and size of orders that could be executed," the report said.

Specific Examples

sure: $P_{3com} > 1.5 P_{palm}$.

- In March 2000, 3-Com sold 5% of its subsidiary Palm in an initial public offering (IPO)
  - after the IPO, 1 share of the 3-Com would effectively include 1.5 shares of Palm
  - after the first day of trading, Palm was at $95
  - in an efficient market, 3-Com should have been at $142, at least!
  - but it was at $82!
  - market value of the 95% of Palm owned by 3-Com was almost $25 billion greater than the market value of 3-Com.
  - as if all of 3-Com's other assets were worth a negative $25 billion

Arbitrage

- One need to borrow the stock in order to short it.
  - Mutual funds, trusts, and asset managers usually lend shares and demand fees.
- It might be hard or impossible to borrow.
- It is costly to borrow shares. If the stock price is $10, and lending fee is 10%, then one need to return one share plus $1 if he borrows the stock for a year.
- Cost of shorting was 35% in July 2000 for Palm.

Superstition Everywhere

- Chang and Du (2022)



1 million USD



Stock primary market

- IPO and PEP

Panel A. A share and GEM, hundredths place

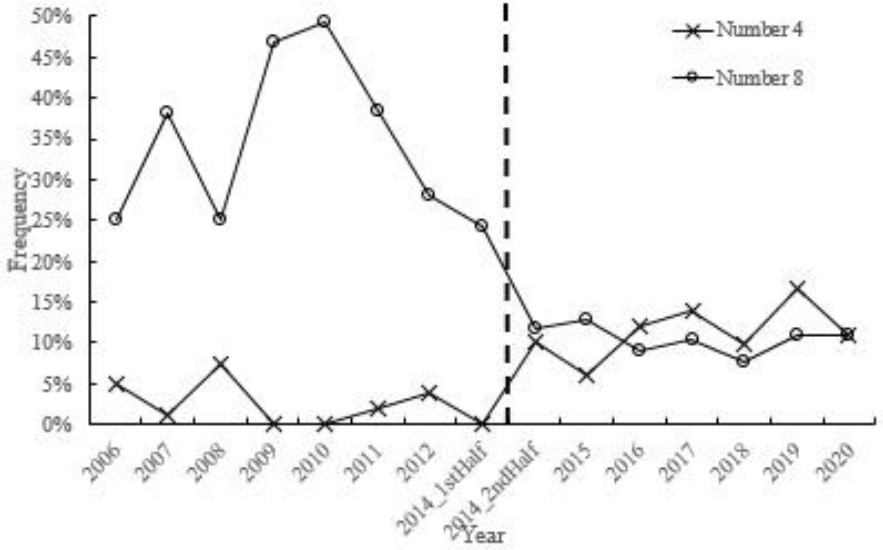
Period: 1/1/1995 —— 6/30/2014										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	57	77	77	48	133	128	55	344	117	1,036
%	5.50	7.43	7.43	4.63	12.84	12.36	5.31	33.20	11.29	100
t-stats	-7.91	-4.51	-4.51	-9.91	1.66	1.22	-8.33	15.09	0.19	
Period: 7/1/2014 —— 3/31/2021										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	129	141	149	150	134	146	156	140	137	1,282
%	10.06	11.00	11.62	11.70	10.45	11.39	12.17	10.92	10.69	100
t-stats	-1.25	-0.13	0.57	0.66	-0.77	0.31	1.16	-0.22	-0.49	

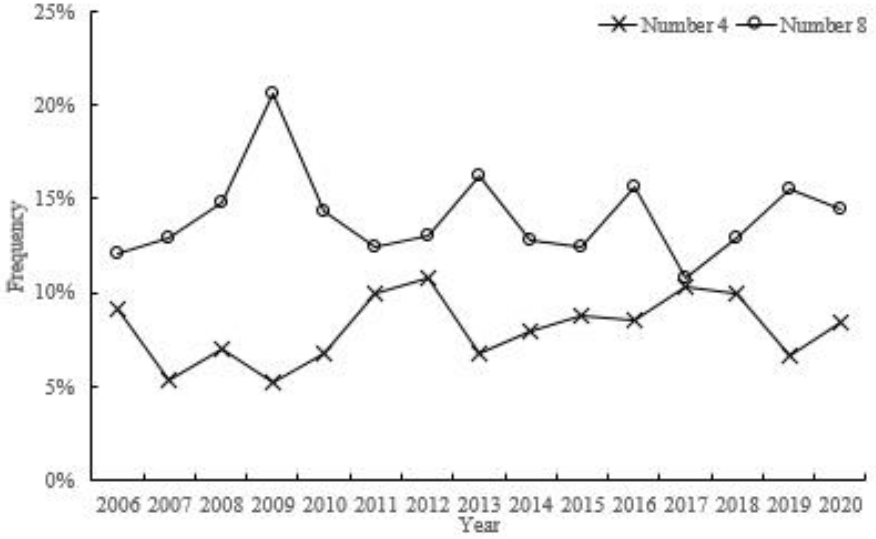
Panel D. Private Equity Placement, Hundredths place

Period: 1/1/1998 —— 3/31/2021										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	493	412	396	306	485	419	344	508	342	3,705
%	13.31	11.12	10.69	8.26	13.09	11.31	9.28	13.71	9.23	100
t-stats	3.93	0.02	-0.83	-6.31	3.57	0.38	-3.83	4.60	-3.95	

Stock primary market

- IPO and PEP: time series analysis





Stock secondary market

- The method to calculate close price changed on July 1, 2006 for Shenzhen Stock Exchange and on August 20, 2018 for Shanghai Exchange changed

Panel A: Open price										
#	1	2	3	4	5	6	7	8	9	Total
Count	775,967	709,964	675,326	586,906	1,063,477	716,495	616,445	927,662	753,483	6,825,725
%	11.37	10.40	9.89	8.56	15.58	10.50	9.03	13.59	11.04	100
t-stats	21.17	-60.75	-110.00	-230.00	321.96	-52.35	-190.00	189.04	-6.02	
Panel B: Close price determined by batched call auction										
#	1	2	3	4	5	6	7	8	9	Total
Count	548,197	562,220	553,613	517,644	702,089	563,382	531,152	647,270	586,936	5,212,503
%	10.52	10.79	10.62	9.93	13.47	10.81	10.19	12.42	11.26	100
t-stats	-44.22	-23.93	-36.33	-90.10	157.71	-22.27	-69.52	90.45	10.77	
Panel C: Close price determined by weighted average										
#	1	2	3	4	5	6	7	8	9	Total
Count	248,575	242,281	239,587	245,035	251,461	240,152	242,940	257,957	273,692	2,241,680
%	11.09	10.81	10.69	10.93	11.22	10.71	10.84	11.51	12.21	100
t-stats	-1.06	-14.62	-20.51	-8.65	5.05	-19.27	-13.18	18.59	50.22	

Stock secondary market

- Time series analysis

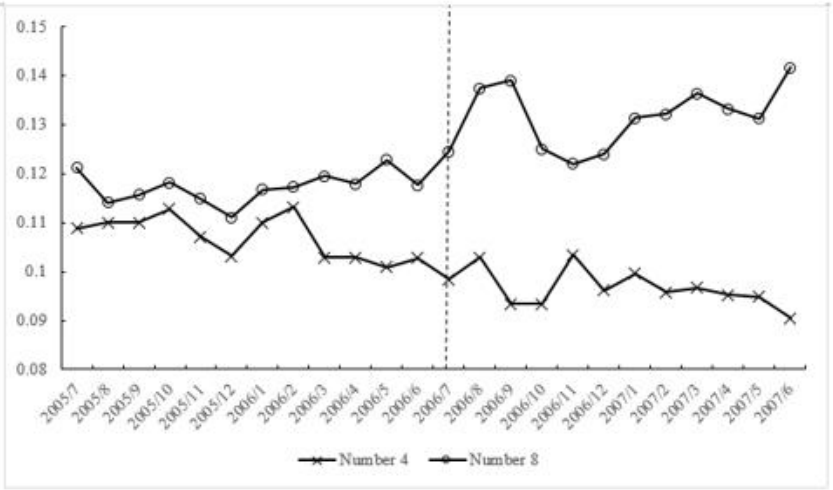


Figure 3.1. The probability of the hundredths digit being 4 & 8 for the close price of stocks listed on the main board at Shenzhen Exchange.

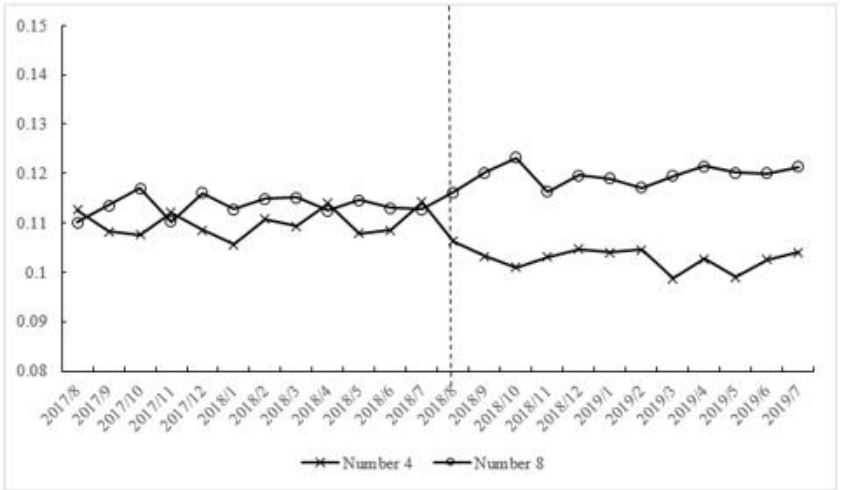


Figure 3.2. Frequency plot of hundredths place digit being 4 & 8 for the closing price of stocks listed on Shanghai Exchange.

FX market

Panel A: Close price USD										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	476	566	515	529	1,095	548	514	686	551	5,480
%	8.69	10.33	9.40	9.65	19.98	10.00	9.38	12.52	10.05	100
t-stats	-6.37	-1.90	-4.35	-3.65	16.42	-2.74	-4.40	3.15	-2.60	
Panel B: Close price EUR										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	430	451	448	443	493	470	398	496	434	4,063
%	10.58	11.10	11.03	10.90	12.13	11.57	9.80	12.21	10.68	100
t-stats	-1.09	-0.02	-0.17	-0.43	2.00	0.91	-2.82	2.13	-0.89	
Panel C: Close price JPY										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	527	502	465	411	631	428	399	539	464	4,366
%	12.07	11.50	10.65	9.41	14.45	9.80	9.14	12.35	10.63	100
t-stats	1.95	0.80	-0.99	-3.84	6.28	-2.91	-4.52	2.48	-1.04	
Panel D: Close price HKD										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	575	613	620	564	728	637	687	751	715	5,890
%	9.76	10.41	10.53	9.58	12.36	10.81	11.66	12.75	12.14	100
t-stats	-3.49	-1.77	-1.46	-4.00	2.91	-0.73	1.32	3.77	2.42	

Commodity futures market

Panel A: Open price										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	5,290	5,601	4,561	4,241	7,892	5,475	4,043	6,670	4,840	48,613
%	10.88	11.52	9.38	8.72	16.23	11.26	8.32	13.72	9.96	100
t-stats	-1.17	2.99	-12.56	-18.41	29.04	1.33	-21.76	16.39	-8.35	
Panel B: Close price										
#	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	Total
Count	5,574	6,465	5,674	6,434	6,365	6,435	5,867	6,726	6,110	55,650
%	10.02	11.62	10.20	11.56	11.44	11.56	10.54	12.09	10.98	100
t-stats	-9.03	3.74	-6.54	3.70	1.88	3.06	-4.53	7.15	-0.71	

Limits to arbitrage

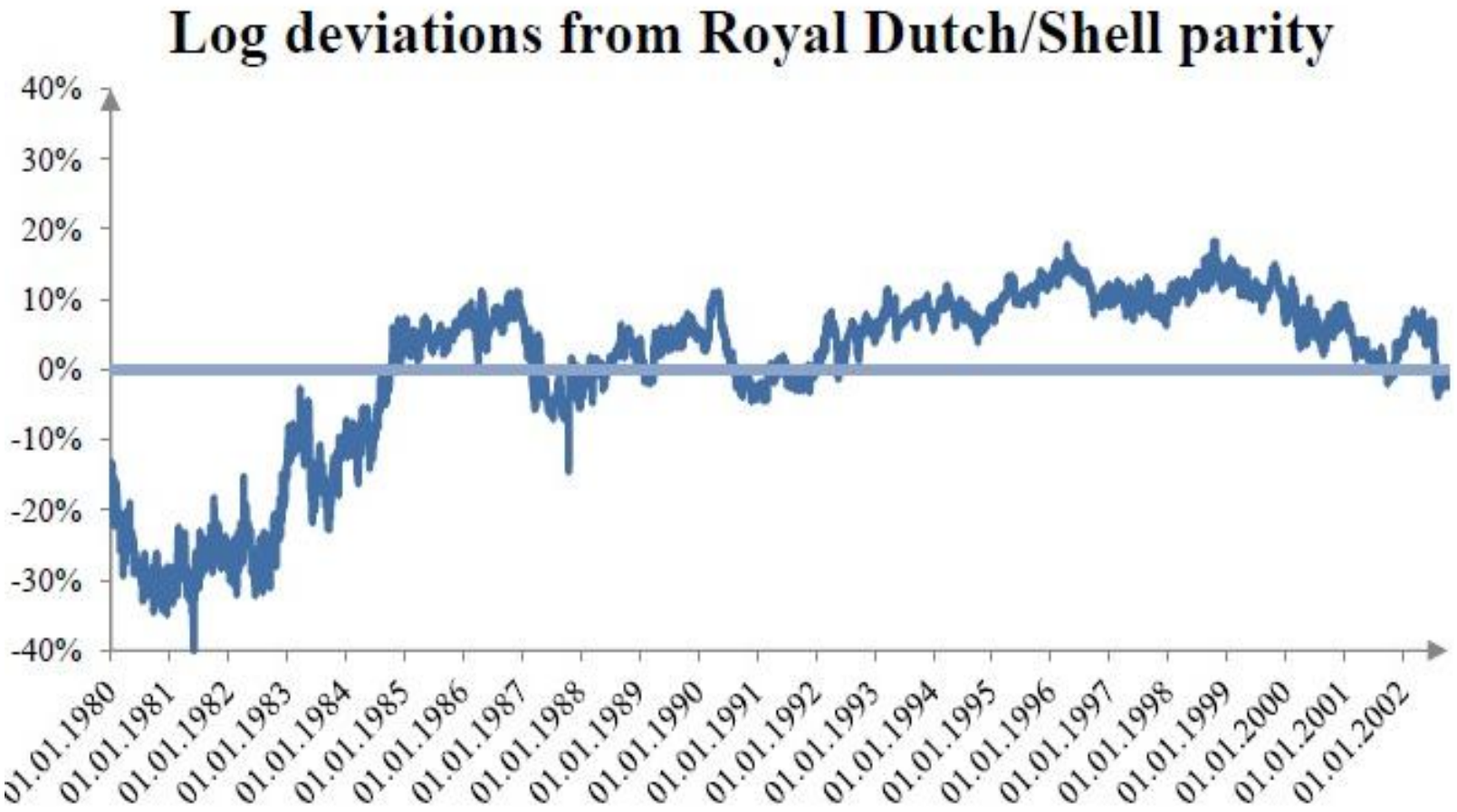
- If stocks are mispriced, why doesn't the arbitrage process eliminate the inefficiency?
  - The risk that mispricing will get worse in the short run
  - Margin Call
  - Fund managers are evaluated frequently
    - The danger is that if the mispricing worsens, even temporarily, investors will withdraw funds
    - Knowing this, the fund managers don't take too aggressive a position against the mispricing, allowing it to survive

Example: Twin shares (Royal Dutch/Shell)

- Twin shares are shares that are claims to the same cash-flow stream
- Royal Dutch shares
  - Trade in the Netherlands and the U.S.
  - Are a claim to 60% of the combined firm's cash flow
- Shell shares
  - trade in the U.K. and the U.S.
  - are a claim to 40% of the combined firm's cash flow

- In an efficient market, must have:
- but

$$\frac{\text{Price(Royal Dutch)}}{\text{Price(Shell)}} = 1.5$$



- prices can stay wrong, or become even more wrong

Summary

- Debate over market efficiency continues …
- Both sides agree that there are many examples of "anomalies"
- Disagree on what those "anomalies" mean for market efficiency
- Even if prices are "wrong", that doesn't mean it's easy to beat the market!
- Next time: do hedge funds, mutual funds, etc. beat the market?